FUTURE STATE JOURNEY MAP

Batch Retrieval for Equipment Review Workflows

Part Identification Repository · Pilot
All research data is simulated

PERSONA
Equipment review engineer, an engineer who verifies parts for equipment reviews and is under delivery pressure. The engineer starts the workflow, which may be built in collaboration with others.

GOAL
Complete a BOM-wide equipment review on schedule without manually looking up each part.

	1 Submit	2 Retrieve	3 Analyze	4 Deliver
USER ACTION	Start the workflow, which submits up to 1,000 part numbers in one authenticated request (target) → No sequential loops	Receive all results in under 1,500 ms (target) → Review starts immediately	Pass lean, UI-free payload to AI agent → ≥75% fewer tokens (target)	Finish BOM-wide equipment review on schedule → Deadline met without workarounds
INTERNAL STATE	Confident the sanctioned path works → No urge to improvise a workaround	Trusts the system is fast → Stays focused on engineering judgment	Empowered, as AI cross-references the BOM → Effort shifts to decisions	Secure and compliant → No insecure CSV dumps to explain
PAIN POINT ADDRESSED	Replaces custom wrappers → No hundreds of sequential GETs	One batch call avoids rate limits → No halfway failures Invalid part numbers are flagged without failing the batch → No full restart after a mid-run failure	Strips layout, image URLs, styling → Ends payload bloat and token waste	Solves the single-lookup limit → Removes the need for scraping data from the legacy UI and CSV dumps Security visibility assumed through existing activity logging and reporting

COMPETITIVE ADVANTAGES OVER THE SHADOW WORKAROUND

Speed
1,000 lookups in under 1,500 ms (target)
→ Replaces hundreds of sequential requests

Security
Sanctioned path
→ Removes the need for shadow workarounds
Visibility assumed through existing activity logging and reporting

Cost
Stripped, AI-ready payloads
→ ≥75% lower AI computing cost (target)

Known risk: The 1,500 ms target for 1,000 parts depends on backend query efficiency. N+1 query degradation (Critical, see Thematic Synthesis) is a known risk to resolve for the pilot and before general production release.

Assumptions: All research data is simulated. The feature is piloted with a select team of engineers vetted for security. Role-based access control, activity logging and reporting are assumed, not built. Full security guardrails are required before general production release.